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Smart People Safety and Crewing tool

A. Antunes and T. Tatagiba, Intcom S.A, Rio De Janeiro, Brazil

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Brazil, a continental-sized nation with over 7,000 kilometers of coastline, hosts one of the most advanced and expansive exploration and production (E&P) sectors in the global energy landscape. The country's offshore operations are concentrated primarily in the southeastern and northeastern basins, with particular emphasis on the prolific pre-salt reserves located in the Santos and Campos Basins. These ultra-deepwater fields have positioned Brazil as a strategic player in global oil production, with pre-salt output currently exceeding 3.3 million barrels per day (bpd)—a figure that continues to grow due to ongoing investments in subsea infrastructure and reservoir development.

As of 2025, Brazil operates more than 48 Floating Production Storage and Offloading (FPSO) units, dozens of offshore drilling rigs, and an estimated fleet of 500 support vessels, including platform supply vessels (PSVs), anchor handling tug supply (AHTS) ships, and crew transfer vessels. This extensive offshore footprint places Brazil among the top five oil-producing nations globally and underscores the complexity of its maritime logistics ecosystem.

The scale and dispersion of these operations present significant logistical challenges, particularly in the coordination of personnel mobilization, cargo transport, emergency response readiness, and compliance with stringent health, safety, and environmental (HSE) standards. Offshore installations are often located hundreds of kilometers from the coast, requiring precise scheduling of helicopter flights, vessel rotations, and crew changes to maintain operational continuity and minimize downtime.

Introduction

Brazil, a continental-sized nation with over 7,000 kilometers of coastline, hosts one of the most advanced and expansive exploration and production (E&P) sectors in the global energy landscape. The country's offshore operations are concentrated primarily in the southeastern and northeastern basins, with particular emphasis on the prolific pre-salt reserves located in the Santos and Campos Basins. These ultra-deepwater fields have positioned Brazil as a strategic player in global oil production, with pre-salt output currently exceeding 3.3 million barrels per day (bpd)—a figure that continues to grow due to ongoing investments in subsea infrastructure and reservoir development.

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Moreover, the dynamic regulatory environment—driven by entities such as ANP (Agência Nacional do Petróleo) and Petrobras—imposes rigorous requirements for workforce certification, local content compliance, and real-time reporting. These factors necessitate the deployment of robust, scalable, and integrated logistics systems capable of managing high-volume personnel flows, ensuring documentation accuracy, and supporting predictive planning across multiple operational nodes.

In this context, digital transformation initiatives—such as AI-driven onboarding platforms, real-time tracking systems, and carbon-optimized transport planning—are increasingly critical to enhancing efficiency, reducing operational risk, and aligning with environmental, social, and governance (ESG) imperatives. The Brazilian offshore sector thus serves as a compelling case study for the intersection of industrial scale, regulatory complexity, and technological innovation in modern energy logistics.

Additionally, Brazil's Equatorial Margin, spanning over 2,200 km along the northern coast, is poised to become a major offshore oil frontier with estimated reserves exceeding 30 billion barrels of oil equivalent. Forecasts suggest production could reach 1.1 million barrels per day by 2029, accounting for roughly one-third of Brazil's current output. Often referred to as the "new pre-salt," the region holds strategic importance for sustaining national oil production beyond 2030, despite challenges related to environmental licensing and deepwater exploration.

One of the greatest challenges faced by the sector in Brazil lies in ensuring the alignment of qualified professionals with the appropriate projects, while maintaining safety and operational efficiency in highly complex environments. Additionally, the industry must navigate a broad array of requirements related to local content, labor regulations, and compliance with Petrobras standards.

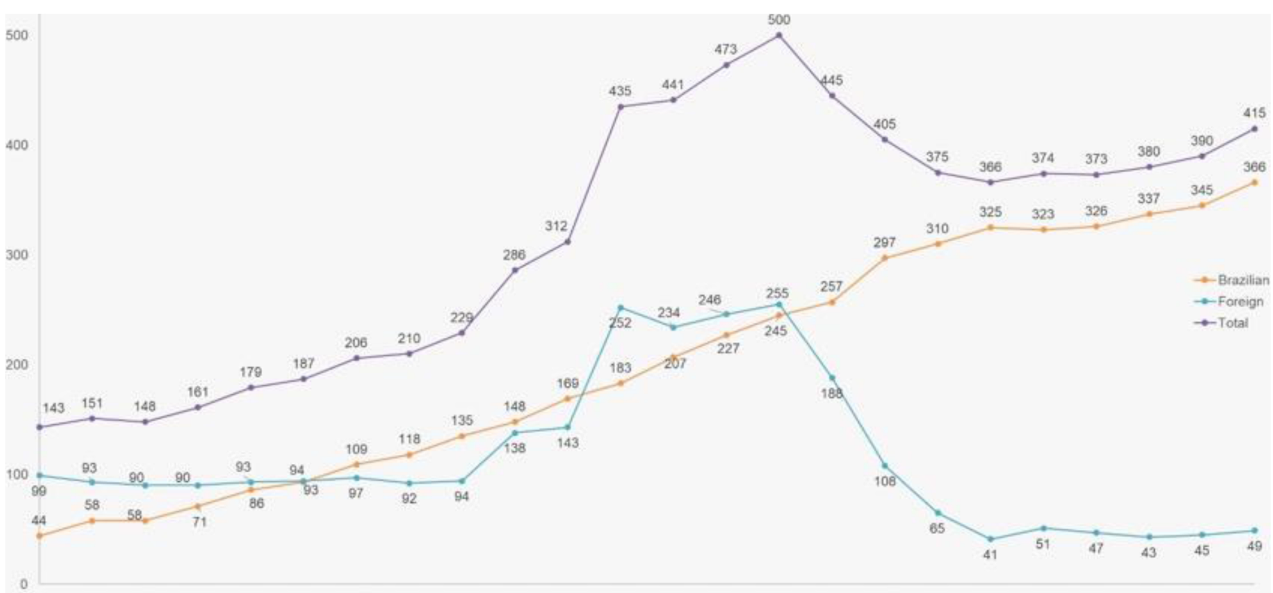

Figure 1-A—OSV Deployment Trends in Brazilian Offshore Fields

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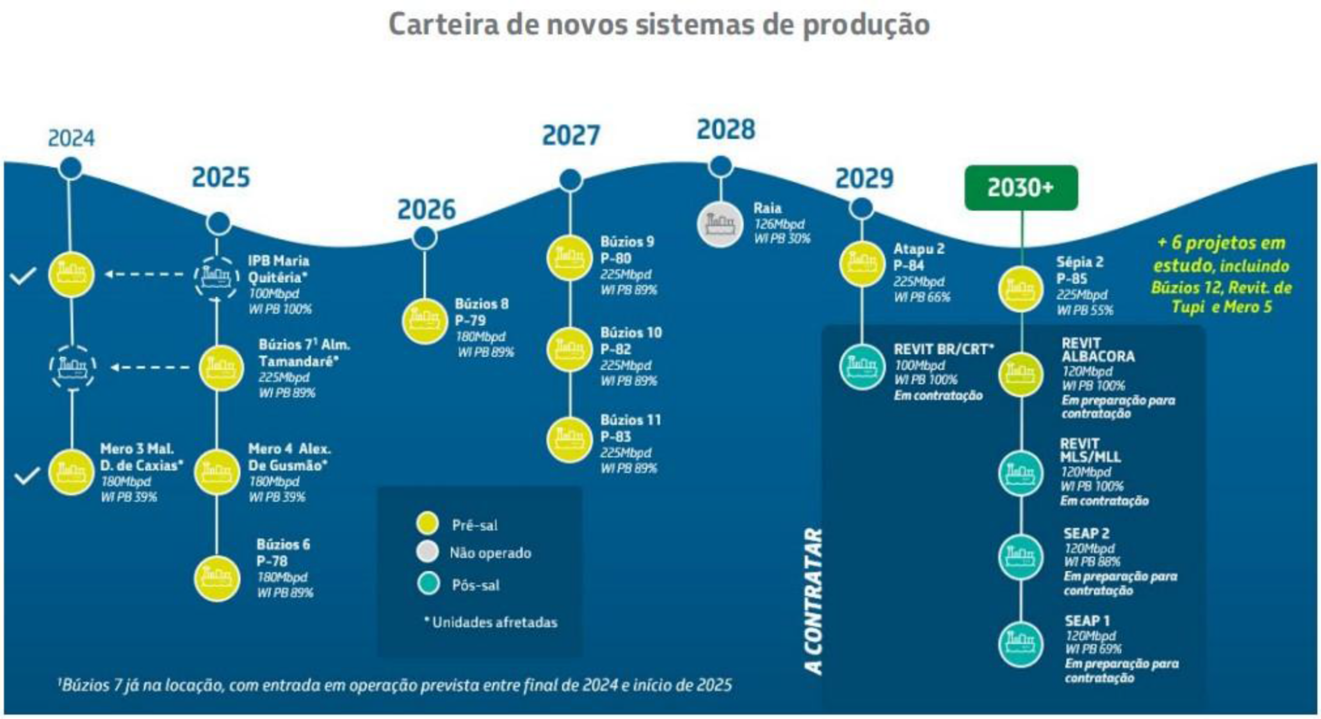

Figure 1-B—FPSO PETROBRAS new projects (2024–2030)

The sector as a whole has faced persistent challenges in simultaneously ensuring operational effectiveness and meeting complex compliance requirements. These constraints have resulted in increased operational costs, heightened risk of human error, fragmented data management, and limited auditability across critical processes.

Accordingly, the objective of this study is to contribute to the advancement of offshore operational efficiency by validating a technology-driven methodology aimed at optimizing crew management, streamlining personnel logistics, and reinforcing human safety protocols within offshore environments.

Methods and Theory

Recent advancements in technologies such as Machine Learning (ML) and Artificial Intelligence (AI) have been increasingly adopted by energy companies to enhance operational safety and efficiency. These technologies also present significant potential for integration within Human Resources (HR) and Health, Safety, and Environment (HSE) departments, offering opportunities to optimize workforce management and strengthen risk mitigation strategies.

Despite recent breakthroughs in Artificial Intelligence—particularly in Generative AI (GenAI) and Large Language Models (LLMs)—their application within the sector remains nascent. While GenAI introduces novel capabilities, it also presents inherent challenges including limited domain-specific training data, model interpretability constraints, scalability bottlenecks, and concerns around reliability and trust. Nevertheless, these technologies represent a promising frontier for augmenting operational forecasting, predictive analytics, computer vision pipelines, and budgetary optimization through advanced data synthesis, contextual inference, and multimodal integration.

Our research, conducted across major oil companies, suppliers, and leading ERP platforms such as SAP and Oracle, indicates that while industry leaders have invested heavily in addressing operational bottlenecks, these efforts have yielded limited success. Concurrently, numerous specialized software solutions exist—many of which are highly effective—but each tends to concentrate on its core functionalities, such as training

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management, payroll, employee records, timesheets, and document handling. Only a select few systems are designed to comprehensively address the full spectrum of processes involved in crewing operations.

Proposed Methodology and Technology

This paper details the research and development of an integrated technological framework and operational methodology for offshore crew management and safety, developed in collaboration with Wilson Sons Offshore, Brazil*. The initiative was benchmarked against five leading oil and gas operators, enabling a comparative analysis that identified systemic inefficiencies and operational bottlenecks—particularly in personnel logistics, embarkation workflows, and onboard safety assurance within offshore environments.

The research enabled the creation of a platform capable of addressing the full lifecycle of offshore operations. It covers efficient crew rotation planning, document compliance, data governance, health, safety and environment (HSE) protocols, onboard safety, and cost management. The solution integrates multiple operational layers into a unified system, improving transparency, compliance, and efficiency.

The platform centralizes personnel data through seamless integration with existing ERP systems, ensuring GDPR-compliant data governance. It automates workflows for travel planning, approvals, and crew scheduling, while offering mobile access to field personnel for greater agility and responsiveness.

Real-time updates enable accurate tracking of training status, certifications, and crew availability. Wearable devices provide continuous location tracking of offshore personnel, enhancing safety and emergency response capabilities. By consolidating passenger movements and optimizing transport routes, the system reduces unnecessary trips, lowering fuel consumption and CO_2 emissions.

Advanced dashboards and analytics support strategic planning, allowing managers to align operational performance with environmental goals and compliance standards.

From a **technology** and **architectural** standpoint, our solution is built upon the Microsoft ecosystem. The backend is developed using the .NET Framework, while data management is supported by SQL Server hosted in the AZURE cloud.

To ensure scalability and operational efficiency, we leverage containerization through Azure Kubernetes Service (AKS). AKS provides a robust platform for deploying and managing containerized applications by abstracting the control plane, thereby enabling teams to concentrate on application development and delivery. Its seamless integration with other Azure services enhances productivity and supports a unified cloud strategy. Additionally, AKS offers advanced capabilities in scalability, security, and cost optimization, making it well-suited for modern enterprise environments.

On the frontend, we utilize Angular and React.js to deliver responsive, high-performance user interfaces.

Automating Document and Training Data Management in Offshore Employee Onboarding Using AI and OCR Technologies

The onboarding process for new employees in offshore operations presents significant logistical and administrative challenges, particularly in the management of personal documentation and training records. This process traditionally involves the manual review and validation of a large volume of documents—including certifications, licenses, medical clearances, and training transcripts—followed by the extraction of relevant data points and their entry into disparate workforce management systems. Such procedures are not only labor-intensive but also prone to human error, delays, and compliance risks.

To address these inefficiencies, we have developed and implemented an artificial intelligence–driven solution that integrates Optical Character Recognition (OCR) and intelligent data validation into the onboarding workflow. The system is designed to automatically ingest scanned or digital documents submitted by prospective offshore personnel, identify and extract structured data fields (e.g., certificate type, expiration date, training completion status), and validate them against predefined compliance criteria.

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The OCR engine is trained on domain-specific document formats commonly used in offshore industries, enabling high-accuracy parsing of diverse layouts and multilingual content. Once extracted, the data is cross-referenced with regulatory and contractual requirements, and then seamlessly synchronized with the organization's human resources and operational planning platforms via secure APIs.

This automation framework yields several measurable benefits:

- **Operational Efficiency**: Reduces document processing time by over 80%, enabling faster onboarding cycles.
- **Data Integrity**: Minimizes manual input errors and ensures consistent compliance with contractual obligations.
- **Scalability**: Supports high-volume onboarding scenarios without proportional increases in administrative overhead.
- **Real-Time Visibility**: Provides immediate access to validated personnel data for shift planning, crew allocation, and audit readiness.

The proof-of-concept (POC) implementation of this system typically spans a three-month period and requires minimal disruption to existing workflows. It is designed to operate in parallel with manual processes during the transition phase, allowing for comparative performance analysis and stakeholder feedback.

This approach represents a significant advancement in digital workforce management for offshore operations, aligning with broader industry trends toward automation, compliance assurance, and data-driven decision-making.

***Wilson Sons Offshore is a key division of Wilson Sons, one of Brazil's oldest and most diversified maritime logistics companies, founded in 1837. The offshore segment plays a critical role in supporting Brazil's oil and gas industry, particularly in pre-salt exploration and production.**

Field Application

The technology was conceived as a Research and Development initiative in partnership with Wilson Sons Offshore, a company operating more than 20 vessels and managing over 1,000 crew members. The operational landscape was fragmented, with multiple disconnected systems and manual controls that lacked integration and centralization.

Petrobras, Brazil's largest oil and gas operator, plays a central role in shaping the regulatory landscape of offshore energy operations in the country. As a state-controlled entity, Petrobras enforces rigorous compliance protocols that extend across contractor selection, workforce qualification, safety standards, and documentation requirements. These protocols are further compounded by Brazil's robust legal framework governing local content obligations, labor protections, and environmental oversight.

Companies operating in this environment must navigate a multilayered set of requirements, including the validation of worker certifications, adherence to union agreements, and fulfillment of contractual clauses related to training, documentation, and shift planning. The enforcement of these standards is not only strict but also dynamic, with frequent updates and audits that demand continuous monitoring and adaptation.

This regulatory complexity has historically posed significant challenges to maintaining real-time visibility over operational costs and compliance status. In many cases, offshore contractors and service providers rely on fragmented systems and manual workflows to manage personnel data, training records, and certification tracking. The absence of centralized data repositories and integrated process automation leads to inefficiencies such as duplicated efforts, delayed onboarding, and increased exposure to non-compliance penalties.

Moreover, the lack of interoperability between HR systems, safety platforms, and operational planning tools impairs decision-making and hinders proactive risk management. These limitations elevate operational

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risk, particularly in high-stakes offshore environments where crew readiness, regulatory adherence, and cost control are tightly interlinked.

Addressing these challenges requires a digital transformation strategy that integrates intelligent data management, real-time compliance monitoring, and scalable automation across the onboarding and workforce lifecycle. Such an approach not only enhances regulatory alignment but also contributes to improved operational resilience and cost efficiency.

The platform was deployed across Wilson Sons Offshore's fleet of 20 vessels, including Anchor Handling Tug Supply (AHTS), Pipe Laying Support Vessels (PLSV), and Platform Supply Vessels (PSV). It enabled end-to-end management of all crew-related processes, including:

- **Recruitment and Onboarding**
  This module streamlines the end-to-end process of attracting, evaluating, and integrating new personnel. It includes:
  - Job requisition and approval workflows
  - Candidate sourcing, screening, and selection tools
  - Digital onboarding checklists and task automation
  - Role-specific induction programs and compliance briefings
  - Integration with background verification and medical clearance systems
- **Document Compliance and Validation**
  Ensures that all workforce documentation meets regulatory and internal standards. Key features include:
  - Centralized repository for certifications, licenses, and IDs
  - Automated expiry alerts and renewal tracking
  - Validation workflows aligned with industry-specific compliance (e.g., HSE, maritime, labor laws)
  - Audit-ready documentation trails and version control
- **Integration with ERP Systems**
  Facilitates seamless data exchange between the workforce management tool and enterprise systems such as SAP, Oracle, or Microsoft Dynamics. Capabilities include:
  - Real-time synchronization of employee records, payroll, and cost centers
  - API-based connectivity for timesheets, financials, and procurement modules
  - Reduction of manual data entry and duplication errors
  - Enhanced visibility across HR, finance, and operations
- **Governance and Training Management**
  Supports regulatory compliance and workforce readiness through structured training programs. Features include:
  - Role-based training matrices and certification tracking
  - Scheduling and enrollment for mandatory and optional courses
  - E-learning integration and performance assessments

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  - Governance dashboards for training compliance and audit reporting

- **Crew Rotation Planning and Scheduling**
  Optimizes crew deployment across operational sites while maintaining compliance and efficiency. Includes:
  - Shift planning, rotation cycles, and leave management
  - Skill-based crew allocation and availability tracking
  - Integration with travel logistics and site access systems
  - Scenario modeling for contingency planning
- **Logistics Coordination and Service Tracking**
  Manages the logistical aspects of workforce mobilization and support services. Functionalities include:
  - Transportation booking and tracking (air, sea, land)
  - Accommodation and catering coordination
  - Service-level monitoring and vendor performance analytics
  - Cost attribution and reconciliation with operational budgets
- **Activity Reporting and Payroll Accuracy**
  Ensures accurate compensation aligned with actual work performed. Key components:
  - Real-time activity logging and timesheet validation
  - Overtime, hazard pay, and allowance calculations
  - Payroll integration with ERP and tax systems
  - Exception handling and audit trails for financial accuracy
- **Transparent Communication Among Stakeholders**
  Facilitates clear and timely information flow across HR, operations, and external partners. Features include:
  - Role-based access to dashboards and reports
  - Notification systems for approvals, alerts, and updates
  - Collaboration tools for incident reporting and feedback
  - Stakeholder engagement metrics and communication logs

Additionally, the platform generated customized reports tailored to Petrobras audit requirements, ensuring regulatory alignment and operational transparency.

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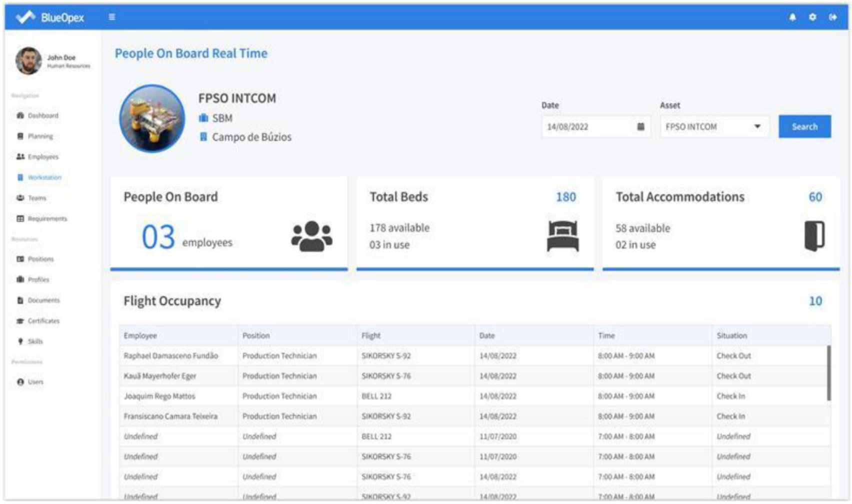

Figure 4-A—Digital real time POB (people on-board)

Results, Observations, Conclusions

The implementation of the platform delivered measurable operational and environmental benefits:

- **Enhanced Safety:** Real-time compliance tracking ensured crew readiness and improved onboard safety.
- **Live Tracking:** Wearables provided continuous location data, improving emergency response capabilities.
- **Data-Driven Insights:** Custom dashboards enabled informed decision-making in resource allocation and logistics planning.
- **Improved Communication:** Mobile access streamlined coordination, enhanced transparency, and accelerated approvals.
- **Seamless Integration:** Compatibility with existing HR and logistics systems ensured minimal disruption and high data accuracy.
- **Operational Efficiency:**
  - Reduction in Manual Processes: Percentage decrease in the use of spreadsheets and manual controls.
  - Process Automation Rate: Proportion of workforce management tasks automated through the new system.
  - Time Saved on Administrative Tasks: Average reduction in hours spent on scheduling, payroll, and compliance tracking.
  - Automation of logistics processes reduced planning time and travel expenses.
- **Cost Optimization**:
  - Reduction in Payroll Errors: Decrease in overpayments or duplicate payments.
  - Avoided Operational Costs: Estimated savings from eliminating redundant tools and inefficiencies.
  - Return on Investment (ROI): Financial return generated from the pilot relative to its cost.
- **Compliance & Data Integrity:**
  - Document Compliance Rate: Percentage of workers with up-to-date and verified documentation.

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- Audit Readiness Score: Improvement in the ability to meet internal and external audit requirements.
- Data Accuracy Rate: Reduction in data discrepancies across workforce records.

Overall, the platform promotes a more sustainable and efficient approach to offshore personnel logistics, aligning operational goals with ESG commitments and industry best practices.

From an **operational** standpoint, the implementation of this solution has yielded measurable improvements across key performance indicators within Wilson Sons' offshore logistics and workforce management processes:

- 50% reduction in travel logistics costs, driven by optimized scheduling and resource allocation
- 80% decrease in time spent managing documents and certifications, enabled by automated validation and centralized digital workflows
- 40% reduction in onboarding time, resulting from streamlined data capture and process automation
- 100% elimination of fines related to missing or expired certifications, through proactive compliance monitoring and alerts
- 80% reduction in costs associated with extra day payments, due to improved planning accuracy and real-time workforce tracking

Novel/Additive Information

This paper presents a comprehensive digital logistics platform designed to address the multifaceted challenges of offshore personnel and asset mobilization. The solution integrates three core technological pillars: enterprise resource planning (ERP) system interoperability, real-time personnel tracking via wearable devices, and carbon-conscious transport optimization algorithms. Together, these components form a unified framework that enhances operational visibility, streamlines logistics coordination, and supports environmental sustainability objectives.

The platform's ERP integration enables seamless data exchange between logistics workflows and corporate systems, ensuring that personnel schedules, asset movements, and compliance records are consistently aligned with broader operational plans. Real-time tracking is achieved through wearable technologies—such as RFID-enabled badges and biometric sensors—that monitor crew location, movement patterns, and boarding status across heliports, vessels, and offshore installations. This data is securely transmitted to centralized dashboards, allowing logistics coordinators to make informed decisions and respond dynamically to schedule changes or safety incidents.

A distinguishing feature of the platform is its embedded carbon optimization engine, which leverages historical transport data, weather forecasts, and crew rotation schedules to recommend routing and modal choices that minimize fuel consumption and greenhouse gas emissions. By quantifying the carbon footprint of each logistical operation, the system empowers operators to make environmentally responsible decisions without compromising service levels or safety standards.

This integrated approach advances the state of knowledge in offshore logistics by demonstrating how digital transformation can be directly linked to measurable reductions in CO_2 emissions. It also provides a scalable and replicable model for oil and gas companies seeking to align their operational practices with environmental, social, and governance (ESG) commitments. The platform supports compliance with international sustainability frameworks, including GHG Protocol reporting and Scope 3 emissions tracking, while enhancing crew safety, reducing costs, and improving scheduling accuracy.

In bridging the gap between operational excellence and ESG accountability, this solution represents a strategic enabler for offshore energy companies navigating the dual imperatives of performance and sustainability in increasingly complex regulatory environments.

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Acknowledgment

We extend our sincere gratitude to Wilson Sons Offshore for their invaluable partnership in developing this technology. Their operational insights, commitment to innovation, and collaborative spirit were fundamental to the success of this project.

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